

A3.1.1 Relational databases



A3.1.1 Explain the features, benefits and limitations of a relational database.

Introduces database fundamentals, emphasizing their role in addressing various challenges by storing data for accessibility, security and widespread use. Students explore data storage, accessibility, security measures and the technology's ability to access global information while protecting private data. The study also covers big data analysis for pattern detection and predictive analysis, alongside machine learning's utilisation of this data to construct models for diverse applications.

Relational Databases: An Overview

A database is an organised collection of data.

In a database the data is organised in tables

Relational databases are a cornerstone of modern data management. They structure data into tables with predefined relationships, offering powerful storage, retrieval, and manipulation tools.

Key features include primary keys for unique identification, foreign keys to link tables, and composite keys for complex identification.

These databases provide numerous benefits, such as minimising data redundancy, ensuring consistency, and supporting concurrent access.

However, they also present limitations, particularly in handling unstructured data, complex hierarchies, and the scalability challenges posed by big data.

Key Features of Relational Databases

Tables: At the heart of a relational database is the table. Data is organised into rows and columns, similar to a spreadsheet. Each row represents a record, and each column represents an attribute or field. This structured format makes it easy to understand and query data.

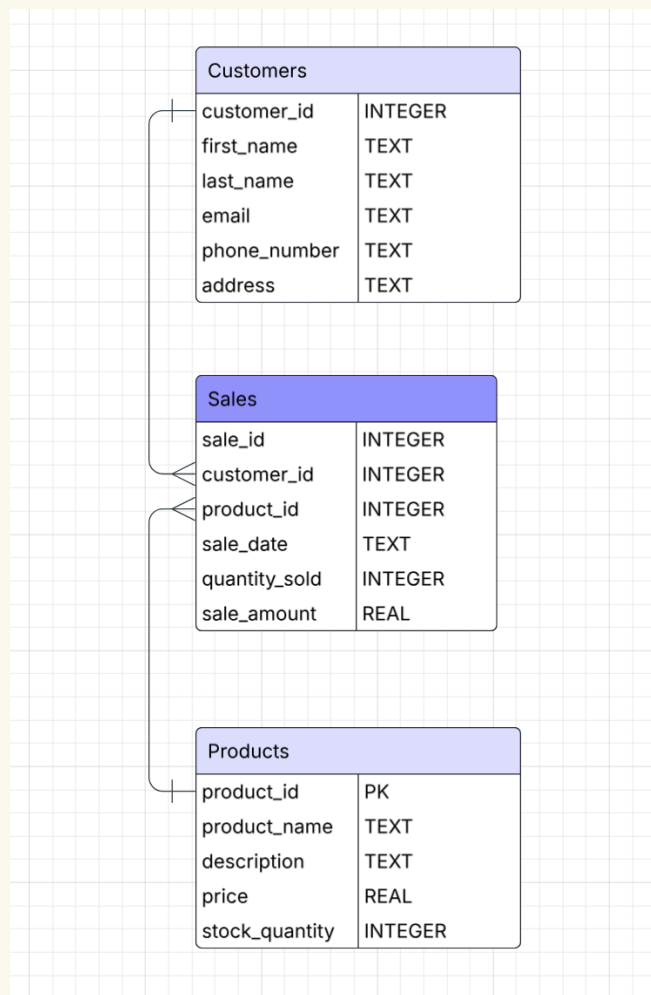
Database Structure Browse Data Edit Pragmas Execute SQL						
Table: Customer						
customer_id	first_name	last_name	email	phone_number	address	
Filter	Filter	Filter	Filter	Filter	Filter	
1	1	Alice	Smith	alice.smith@email.com	123-456-7890	123 Main St, Anytown
2	2	Bob	Johnson	bob.johnson@email.com	987-654-3210	456 Oak Ave, Somecity
3	3	Charlie	Williams	charlie.williams@email.com	555-123-4567	789 Pine Ln, Otherville

Primary Keys: A primary key is a column or a set of columns uniquely identifying each row in a table. No two rows can have the same primary key value. This uniqueness is crucial for maintaining the integrity of the data and for quickly accessing specific records.

Foreign Keys: Foreign keys establish relationships between tables. A foreign key in one table references a primary key in another table, creating a link that allows related data to be retrieved or updated across multiple tables.

Composite Keys: In some cases, a single column cannot uniquely identify rows. A composite key is a combination of two or more columns that together form a primary key. This is used when the uniqueness of a record depends on a combination of values.

Relationships: Relationships are the associations between tables, established through primary and foreign keys.



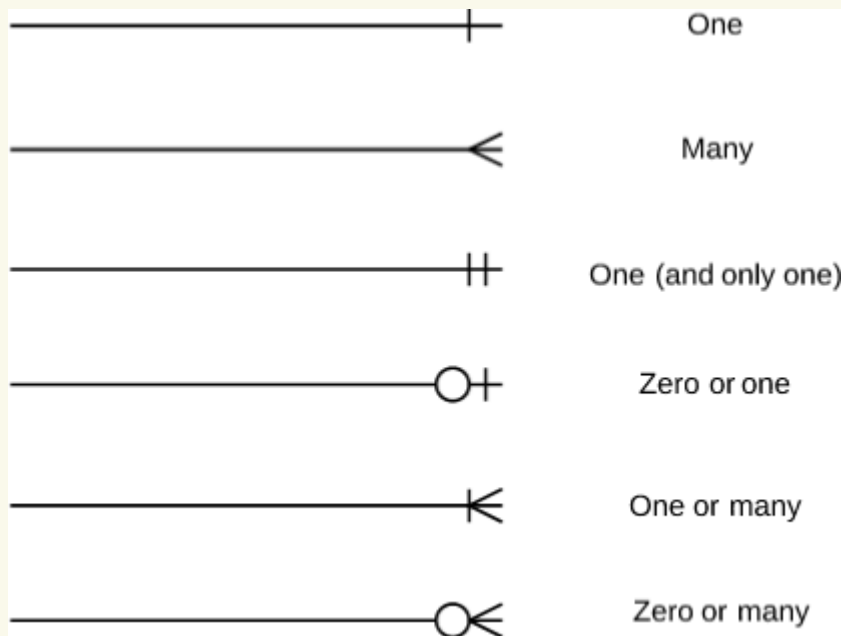
Entity Relationship diagram for simple sales database

Common types of relationships include:

One-to-one: Each record in one table is related to one record in another table.

One-to-many: One record in a table can be related to multiple records in another table.

Many-to-many: Multiple records in one table can be related to multiple records in another table (often involving a junction table).



Benefits of Relational Databases

Data Retrieval: Relational databases excel at retrieving specific data quickly and efficiently.

Structured Query Language (SQL) is the standard language for interacting with relational databases, allowing users to write complex queries to filter, sort, and aggregate data.

Reduced Data Duplication and Redundancy: Normalization is a database design technique that minimizes data duplication. By storing each piece of data in only one place, relational databases reduce redundancy, saving storage space and ensuring consistency.

Data Consistency and Integrity: Relational databases enforce constraints and rules to maintain data consistency and integrity. These may include data type validation, uniqueness constraints, and referential integrity (ensuring that foreign key values are valid).

Concurrency Control: Relational databases manage concurrent access by multiple users. They use techniques like locking to prevent conflicts when multiple users try to modify the same data simultaneously, ensuring that transactions are processed correctly.

Reliable Transaction Processing: Transactions are a sequence of operations that are treated as a single unit of work. Relational databases ensure that transactions are processed reliably, even in the event of system failures, using ACID properties (Atomicity, Consistency, Isolation, Durability).

Scalability: Relational databases can handle growing amounts of data and users. They can be scaled vertically (by adding more resources to a server) or horizontally (by distributing the database across multiple servers), though horizontal scaling can present challenges.

Security Features: Relational databases provide various security features to protect data. These include user authentication, access control (granting or revoking privileges), and encryption to secure data at rest and in transit.

Community Support: Relational databases have a large and active community. This extensive support network means there are plenty of resources, tools, and expertise available to help users design, implement, and maintain relational database systems.

Limitations of Relational Databases

Design Complexity: Designing an efficient and effective relational database can be complex. It requires careful planning to normalize data, define relationships, and optimize queries. Poor design can lead to performance issues and data integrity problems.

Rigid Schema: Relational databases have a rigid schema, which means the structure of the database must be defined in advance. Changes to the schema can be difficult and time-consuming, making it challenging to adapt to evolving data requirements.

Object-Relational Impedance Mismatch: Relational databases and object-oriented programming languages represent data in different ways. This can lead to complexities when mapping objects to database tables, a problem known as the object-relational impedance mismatch.

Hierarchical Data Handling: While relational databases can represent hierarchical data, they are not ideally suited for it. Recursive queries or complex joins may be required to navigate hierarchical structures, which can be inefficient.

Unstructured Data Handling: Relational databases are designed for structured data (data that fits neatly into tables). They are not well-suited for handling unstructured data, such as text documents, images, and videos, which require different types of database systems.

“Big Data” Scalability Issues: While relational databases can scale, they can face challenges when dealing with massive datasets and high-velocity data streams characteristic of "big data."

Distributed database systems and NoSQL databases are often better choices for these scenarios.

A 3.1.1 Quiz

Test your knowledge by trying this Quiz

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structured unstructured hierarchies A primary key

One-to-many ACID properties (Atomicity, Consistency, Isolation, Durability) column

foreign keys Normalization complex composite keys

Structured Query Language (SQL) row primary keys tables Foreign keys

object-oriented programming languages videos data Transactions community

big data

Relational databases are a cornerstone of modern data management. They structure data into **[tables]** with predefined relationships, offering powerful tools for storage, retrieval, and manipulation. Key features include **[primary keys]** for unique identification, **[foreign keys]** to link tables, and **[composite keys]** for complex identification. These databases provide numerous benefits, such as minimizing data redundancy, ensuring consistency, and supporting concurrent access. However, they also present limitations, particularly in handling unstructured **[data]**, complex **[hierarchies]**, and the scalability challenges posed by big data.

At the heart of a relational database is the table. Data is organized into rows and columns, similar to a spreadsheet. Each **[row]** represents a record, and each **[column]** represents an attribute or field. **[A primary key]** is a column or a set of columns that uniquely identifies each row in a table. **[Foreign keys]** establish relationships between tables. A composite key is a combination of two or more columns that together form a primary key.

Relationships are the associations between tables, established through primary and foreign keys. Common types of relationships include One-to-one, [redacted] **[One-to-many]** , and Many-to-many.

Relational databases excel at retrieving specific data quickly and efficiently.

[redacted] **[Structured Query Language (SQL)]** is the standard language for interacting with relational databases. [redacted]

[Normalization] is a database design technique that minimizes data duplication. Relational databases enforce constraints and rules to maintain data consistency and integrity. Relational databases manage concurrency control by multiple users. [redacted]

[Transactions] are a sequence of operations that are treated as a single unit of work. Relational databases ensure that transactions are processed reliably, even in the event of system failures, using [redacted] **[ACID properties (Atomicity, Consistency, Isolation, Durability)]** . Relational databases can handle growing amounts of data and users and can be scaled vertically or horizontally. Relational databases provide various security features to protect data, including user authentication, access control, and encryption. Relational databases have a large and active [redacted] **[community]** .

Designing an efficient and effective relational database can be [redacted] **[complex]** . Relational databases have a rigid schema, which means the structure of the database must be defined in advance. Relational databases and [redacted] **[object-oriented programming languages]** represent data in different ways, which can lead to complexities known as object-relational impedance mismatch. While relational databases can represent hierarchical data, they are not ideally suited for it. Relational databases are designed for [redacted] **[structured]** data and are not well-suited for handling [redacted] **[unstructured]** data, such as text documents, images, and [redacted] **[videos]** . While relational databases can scale, they can face challenges when dealing with massive datasets and high-velocity data streams characteristic of [redacted] **[big data]** .